Day-Ahead Load Forecasting Under Weather Forecast Uncertainty

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Abstract

This report presents a day-ahead electricity load forecasting framework with an emphasis on model robustness under temperature forecast uncertainty. In operational settings, real-time weather information is not exactly known ahead of time and must be forecast, creating a mismatch between model evaluations based on observed temperatures and real-world deployment conditions. To address this, Monte Carlo perturbations are applied to temperature inputs to simulate forecast uncertainty, and performance degradation is quantified using root mean squared error (RMSE). The objective is to evaluate how forecasting models behave under realistic constraints where the true future temperature and recent load information are unavailable at prediction time.

1 Overview

1.1 Problem Definition

Accurate day-ahead load forecasting is essential for grid stability, generation scheduling, and reserve planning. In operational practice, load forecasts rely on weather forecasts rather than realized temperature, introducing input uncertainty that may be overlooked if models are assessed using observed temperature, potentially overstating real-world performance.

This project evaluates not only baseline day-ahead forecasting accuracy, but also how model performance degrades under temperature forecast uncertainty. The objective is to compare models based on both predictive accuracy and robustness under imperfect exogenous inputs.

1.2 Approach

A nowcasting model was first developed to screen features and identify stable predictors. The final day-ahead framework evaluated three model classes: Linear Regression, XGBoost, and a hybrid Linear + XGBoost residual approach.

To reflect operational constraints:

- Short-horizon lag features (e.g., $t-1$ load) were excluded.
- Models were trained using observed temperatures but evaluated using Monte Carlo-perturbed temperature inputs to simulate forecast uncertainty.

Performance was assessed using RMSE over all hours and separately over the top 10% highest load hours. Robustness was quantified by measuring how RMSE increased as temperature forecast uncertainty grew.

1.3 Key Findings

- Under observed temperature inputs ($\sigma = 0$), the hybrid Linear + XGBoost residual model achieved the lowest test RMSE (147.8 MW), outperforming XGBoost (155.9 MW) and Linear Regression (156.9 MW).

- During the top 10% highest load hours, the hybrid model remained the most accurate, while XGBoost exhibited substantially higher peak-hour error, indicating weaker performance in high-demand regimes.

- As temperature forecast uncertainty increased, model rankings shifted. The hybrid model maintained the lowest error at low uncertainty levels but degraded more rapidly than Linear Regression as temperature perturbations increased.

- At higher uncertainty levels (approximately $\sigma \geq 3$°F), Linear Regression became the most robust model, demonstrating smaller relative RMSE increases compared to more flexible models.

- These results reveal a trade-off between baseline accuracy and robustness: models with greater flexibility achieve lower deterministic error but are more sensitive to temperature forecast noise.

2 Data Description

2.1 Dataset Source and Context

The dataset was obtained from the 2025 PG&E Energy Analytics Challenge and consists of historical electricity load and weather data for the San Diego, California region. The full dataset includes three years of hourly observations of load, temperature, and global horizontal irradiance (GHI).

2.2 Load Data

The load data contains hourly aggregated electricity demand (MW) for the San Diego area. The dataset spans 2020–2022, covering multiple seasonal cycles and peak demand events. The hourly resolution enables modeling of intraday patterns, weekly seasonality, and extreme load periods that are critical for operational forecasting.

2.3 Weather Data

The dataset includes time-aligned hourly weather observations from five neighboring sites within the San Diego region. Available weather variables include temperature (°C) and global horizontal irradiance (GHI) recorded at each of the five sites. Thus, the complete dataset consists of hourly system load, five-site temperature measurements, and five-site GHI measurements over the 2020–2022 period.

Although the dataset provides both temperature and global horizontal irradiance (GHI) at five neighboring sites, only temperature was used in the final models of this study. This choice was intentional, as the primary objective of the study is to evaluate model robustness under temperature forecast uncertainty. Including GHI would have required additional assumptions about solar forecast errors and introduced a second source of exogenous uncertainty, which would confound the attribution of performance degradation specifically to temperature uncertainty. Therefore, the feature set was restricted to temperature to isolate and clearly interpret the impact of weather forecast uncertainty on day-ahead load forecasting performance.

2.4 Feature Engineering

A broad set of candidate features was initially constructed, including calendar variables, temperature transformations, lagged load terms, and rolling statistics. To identify a parsimonious and stable feature set, feature screening was conducted using a linear regression nowcast model as a baseline.

Features were iteratively evaluated based on their impact on out-of-sample RMSE on the training period only, and features that did not provide consistent performance improvements were removed. This ensured that feature selection did not introduce information leakage from the test set.

The final retained features include cyclical hour encodings, weekend and notable-day indicators, averaged temperature across the five neighboring sites, a rolling temperature average over the last 6 hours, and temperature-derived transformations such as cooling and heating degree terms. Lagged load features at 24-hour and 48-hour horizons were also included due to their strong predictive value for daily load patterns.

For the day-ahead forecasting setting, the feature set was further restricted to respect operational information constraints. In particular, short-horizon lag features (e.g., $t-1$ load) were excluded, since such information would not be available at the time of day-ahead prediction. Only features that would realistically be known prior to the prediction horizon, or features that would be forecast or derived from forecasts (temperature and its aggregates or transformations), were retained, ensuring evaluation under realistic deployment conditions.

3 Modeling Approach

3.1 Model Classes and Hourly Modeling

Three primary model families were evaluated: Linear Regression, XGBoost, and a hybrid Linear + XGBoost residual approach. For each family, 24 separate models were trained, one for each hour of the day. This hourly modeling strategy was chosen to capture hour-specific load dynamics, as electricity demand exhibits distinct behavior during morning ramp-up, midday and evening peaks, and overnight periods. Training independent models for each hour allows coefficients and nonlinear patterns to adapt to these intraday regimes rather than forcing a single model to fit all hours.

For XGBoost, two hyperparameter strategies were applied. In the hour-specific approach, each of the 24 hourly models was tuned independently using grid search over tree depth, number of estimators, learning rate, and the minimum child weight. In the uniform hyperparameter approach, a single hyperparameter configuration was chosen based on the hour with the highest baseline RMSE (hour 11) and applied to all hourly models. This reduces tuning variance and enforces consistency across hours while maintaining competitive performance.

The same hour-specific versus uniform hyperparameter strategies were applied to the XGBoost component of the hybrid Linear + XGBoost residual model. In this hybrid framework, a linear model first captures the dominant linear and seasonal structure for each hour, and XGBoost is subsequently trained on the resulting residuals to learn nonlinear patterns not explained by the linear model. This two-stage formulation allows the linear model to represent primary structural load drivers while the residual learner focuses on higher-complexity nonlinear corrections.

3.2 Training and Evaluation Setup

Models were trained on data from 2020–2021 and evaluated on a fully held-out test set from 2022. Within the training period, time-series cross-validation was implemented using a five-split expanding-window scheme to preserve chronological ordering and avoid information leakage. In each split, the training window began on January 1, 2020 and expanded forward in time, followed by a contiguous validation window of approximately three months.

For every split, all models were refit from scratch to ensure a fair comparison and to reflect realistic model retraining. Linear Regression models were fit using ordinary least squares within each cross-validation split. For the hourly XGBoost and hybrid Linear + XGBoost models, hyperparameters were tuned separately for each hour using cross-validated RMSE on the corresponding hour-specific subsets. In the uniform hyperparameter variant, the configuration achieving the lowest cross-validated RMSE at hour 11 was selected and then applied unchanged to all 24 hourly models. Feature selection and all model development decisions were performed using the training period only.

Due to the hourly modeling framework, the dataset was first partitioned into 24 hour-specific subsets. Each hourly model was trained and validated on its corresponding subset within each cross-validation split. Predictions from the 24 hourly models were then recombined to reconstruct the full chronological load series, and RMSE was computed on the reassembled validation set.

For the hybrid models, the residual learners were trained within each cross-validation split using out-of-sample residuals generated via the walk-forward procedure described in Section 3.5. Temperature uncertainty analysis was conducted only on the final 2022 test set after model selection was complete, ensuring that robustness evaluation did not influence model tuning.

Model performance was evaluated primarily using root mean squared error (RMSE) over all hours. Model comparison and final selection across model classes were based exclusively on cross-validated global RMSE. As a secondary diagnostic metric, RMSE was also computed on the top 10% highest load hours to assess performance during peak demand periods; this metric was used only for evaluation and not for model selection.

3.3 Monte Carlo Simulation for Temperature Uncertainty

Model robustness to temperature forecast uncertainty was assessed using a Monte Carlo simulation. For each model on the 2022 test set, 100 synthetic temperature series were generated. At each hour, the synthetic forecast temperature was generated recursively from the previous hour's forecast, incorporating (i) an autoregressive drift component centered on the realized temperature trajectory, treated as the latent true path around which unbiased forecast errors evolve; and (ii) an additive Gaussian error term. The standard deviation of this error term was varied between 1–5°F to represent different levels of forecast uncertainty. For each synthetic series, all day-ahead features were recalculated (e.g., 6-hour rolling average, cooling/heating degree terms), and load predictions were generated. RMSE was computed for each simulation, and the mean and standard deviation of RMSE across simulations were recorded for each uncertainty level. A baseline using the actual future temperatures was also included as a reference. This procedure quantifies how model performance degrades as temperature forecast uncertainty increases while preserving realistic temporal correlation in the forecasts.

3.4 Conditional Performance Evaluation

In addition to overall RMSE, model performance was evaluated on several conditional subsets of the 2022 test set, including the top 10% of highest load hours, seasonal partitions (summer and winter), and hourly segments. The Monte Carlo temperature perturbation procedure was first applied to the full test set to generate synthetic temperature trajectories and corresponding load predictions. The resulting predictions were then filtered according to each condition of interest, and RMSE was computed on the filtered subsets.

For each model and each level of temperature uncertainty, 100 simulations were run, and the mean and standard deviation of RMSE across simulations were recorded for every subset. These conditional metrics were used solely for post hoc model comparison and robustness analysis and were not used in any stage of model training, hyperparameter tuning, or model selection.

3.5 Hybrid Residual Modeling Procedure

In the hybrid Linear + XGBoost residual framework, a separate linear regression model was trained for each of the 24 hours of the day using the same hourly partitioning described above. Out-of-sample residuals for XGBoost training were generated using a walk-forward procedure within the training data: for each hour and each cross-validation split, the linear model was refit on progressively expanding training windows and residuals were computed on the subsequent validation segments. These strictly out-of-sample residuals were then used as the target for the XGBoost models. XGBoost was trained either with hour-specific hyperparameters optimized independently for each hour or with a uniform configuration derived from the hour with the highest baseline RMSE (hour 12). Final hybrid predictions were obtained by adding the XGBoost residual predictions to the corresponding linear model forecasts for each hour.

4 Results

4.1 Model Degradation Under Weather Uncertainty

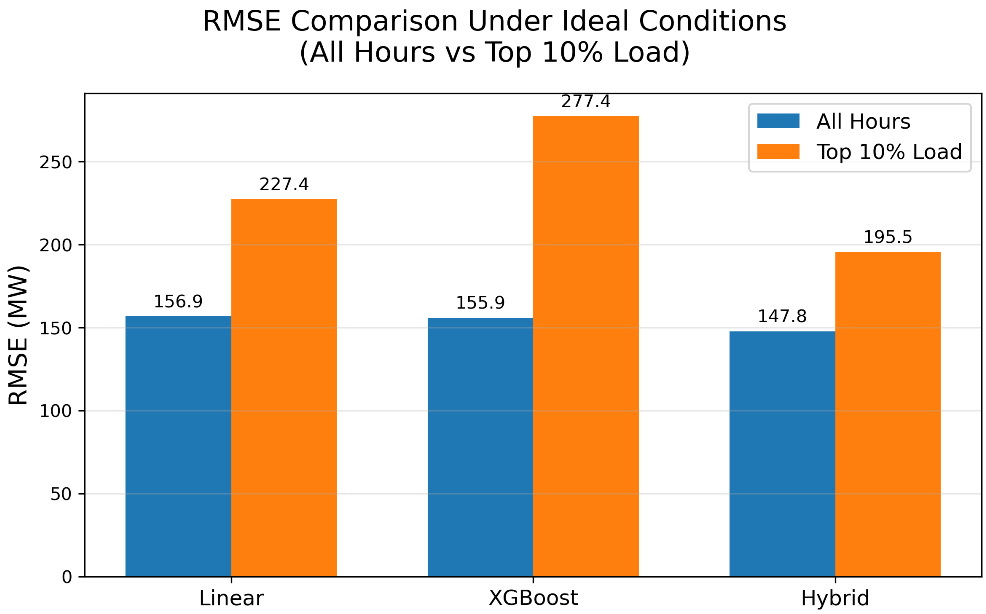

Figure 1: Comparison of test RMSE across model classes on the 2022 test set using observed temperature inputs ($\sigma = 0$). Bars report RMSE over all hours and separately over the top 10% highest load hours. Models shown include Linear Regression, hour-specific XGBoost (per-hour tuned hyperparameters), and the hour-specific Linear + XGBoost residual hybrid model.

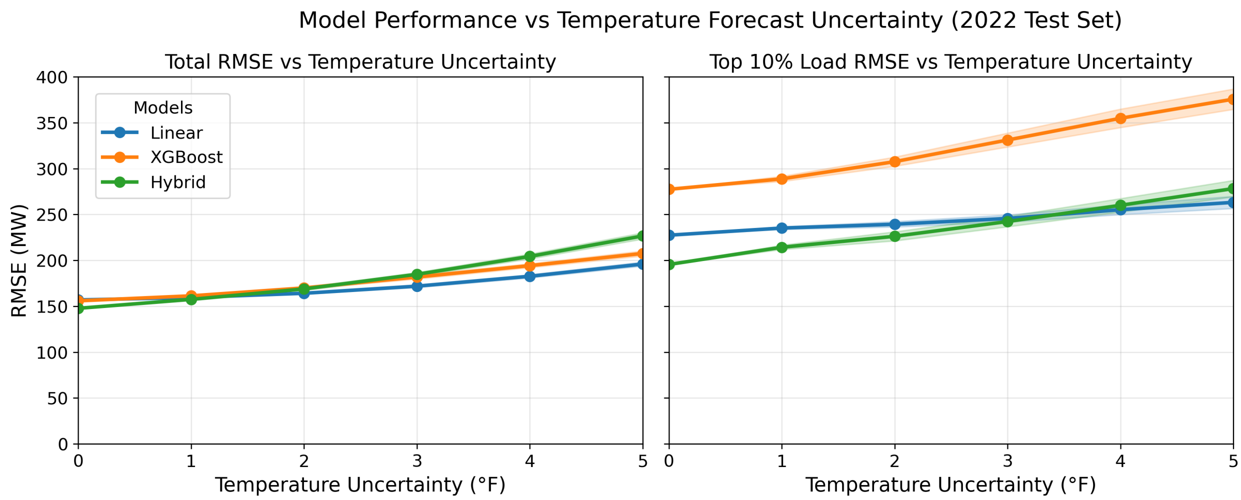

Figure 2: Sensitivity of model RMSE to temperature forecast uncertainty on the 2022 test set. The left panel shows RMSE over all hours, and the right panel shows RMSE over the top 10% highest load hours, both as a function of temperature perturbation standard deviation ($\sigma = 0$–5°F). Curves correspond to Linear Regression, hour-specific XGBoost, and the hour-specific Linear + XGBoost residual hybrid model. Shaded bands denote the standard deviation of RMSE across 100 Monte Carlo simulations, with $\sigma = 0$ representing the observed-temperature baseline.

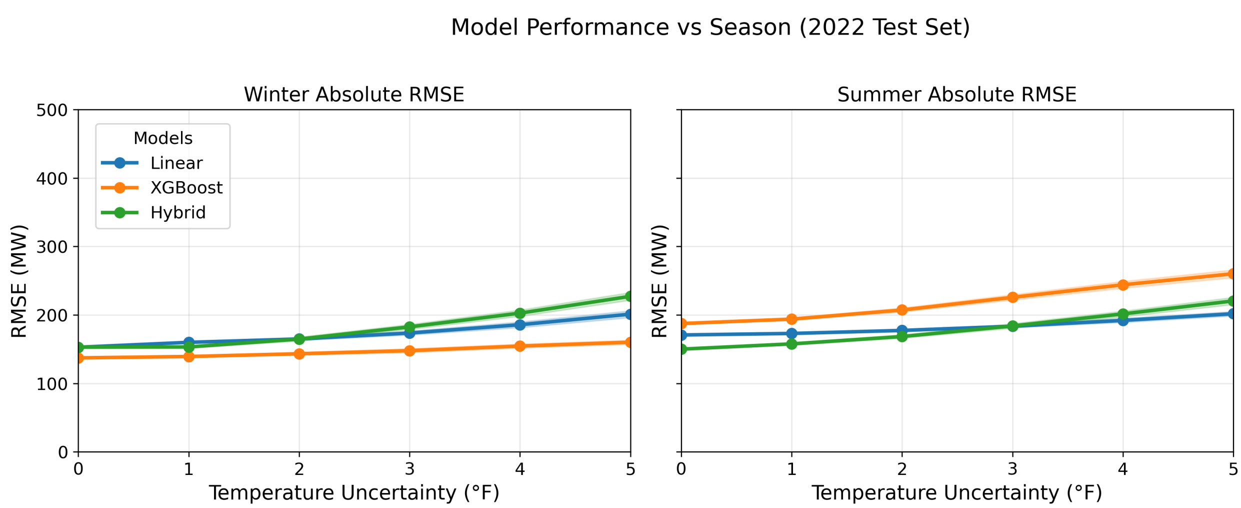

Figure 3: Sensitivity of model RMSE to temperature forecast uncertainty on the 2022 test set, evaluated separately for Winter (November–February) and Summer (June–September). Curves show Linear Regression, hour-specific XGBoost, and the hour-specific Linear + XGBoost residual hybrid as temperature noise ($\sigma = 0$–5°F) increases. Shaded regions denote the standard deviation of RMSE across 100 Monte Carlo simulations.

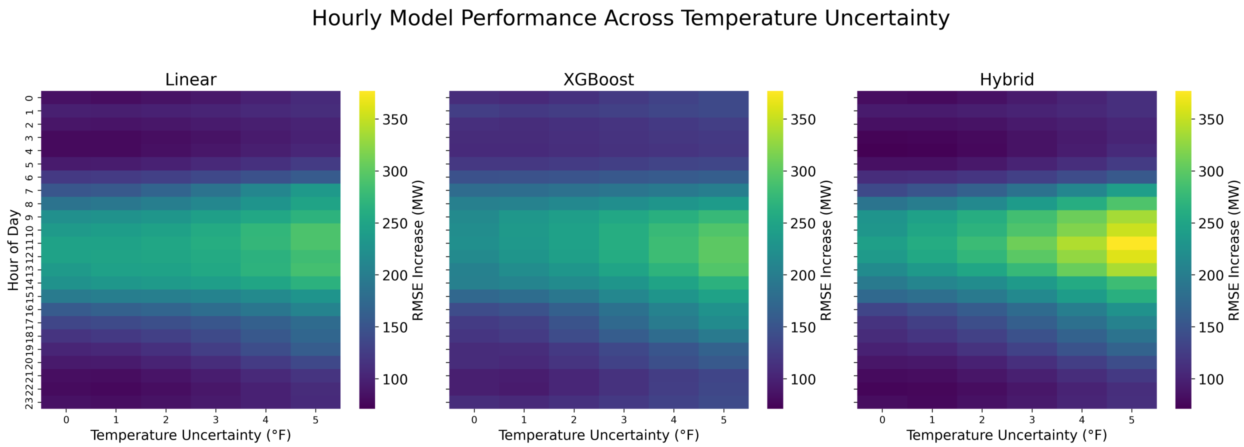

Figure 4: Hourly RMSE as a function of temperature forecast uncertainty (σ = 0–5°F) for each model class on the 2022 test set. Panels show Linear Regression, hour-specific XGBoost, and hour-specific Linear + XGBoost residual hybrid models. Colors indicate the RMSE for each hour and temperature uncertainty combination, highlighting intraday error patterns and model sensitivity to increasing temperature noise.

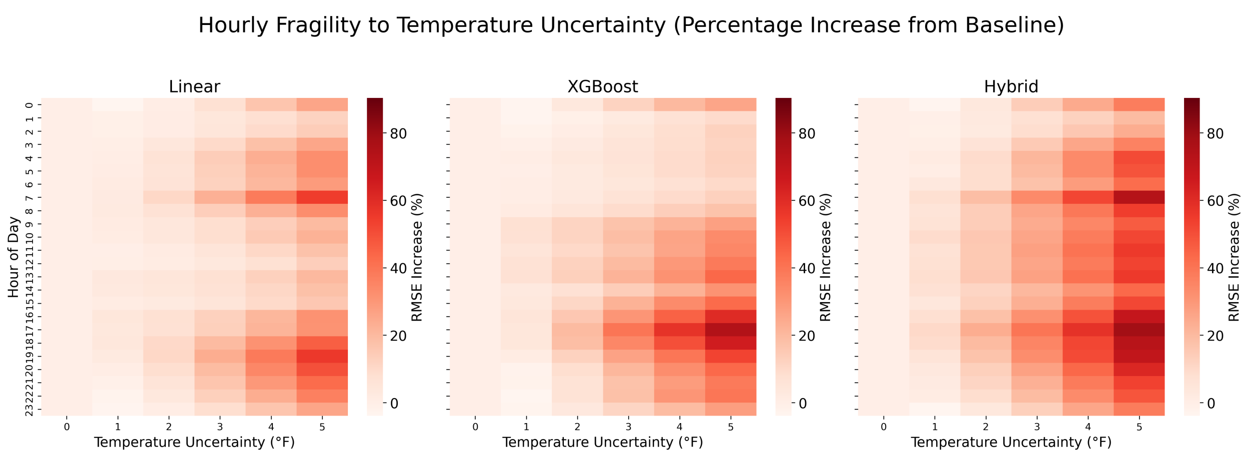

Figure 5: Hourly percentage increase in RMSE relative to the baseline ($\sigma = 0$) as temperature forecast uncertainty grows (σ = 0–5°F) for each model class on the 2022 test set. Panels show Linear, hour-specific XGBoost, and hour-specific Linear + XGBoost residual models. Colors indicate the magnitude of performance degradation as a percentage, highlighting which hours are most sensitive to temperature forecast noise.

5 Discussion

We first examine deterministic model performance under observed temperature inputs before analyzing robustness to forecast uncertainty. This separation clarifies the distinction between baseline predictive accuracy and real-world reliability under imperfect exogenous inputs.

5.1 Overall Model Performance

Using observed temperatures ($\sigma = 0$), the hybrid Linear + XGBoost residual model achieves the lowest RMSE (147.8 MW), followed by XGBoost (155.9 MW) and Linear Regression (156.9 MW). This indicates that the hybrid model effectively captures both linear trends and hour-specific nonlinearities, providing the best baseline accuracy across all hours.

When focusing on the top 10% highest load hours, the hybrid model remains the most accurate, but RMSE increases by roughly 50 MW due to higher variability in peak demand periods. Linear Regression experiences a larger increase (70 MW), while XGBoost performs worst at peaks, with RMSE rising to 120 MW. This suggests that complex nonlinear models like XGBoost can struggle during extreme load conditions if not combined with linear structure. In pure XGBoost, the model must learn both the main trend and nonlinear effects simultaneously, which is more challenging with limited data per hour, particularly for rare high-load events.

5.2 Impact of Temperature Forecast Uncertainty

Introducing temperature forecast noise ($\sigma = 1$–5°F) alters the performance ranking. Across all hours, Linear Regression degrades modestly, whereas both XGBoost and the hybrid model exhibit sharper RMSE increases. The hybrid model shows the largest overall degradation, reflecting its sensitivity to imperfect temperature inputs.

For the top 10% load hours, XGBoost consistently underperforms at all uncertainty levels. The hybrid model initially maintains an advantage, but beyond $\sigma \approx 3$°F, Linear Regression becomes more robust. These trends illustrate the trade-off between baseline accuracy and robustness: more flexible models achieve lower error under perfect inputs but are more sensitive to temperature forecast noise.

From an operational perspective, these findings suggest that selecting models solely based on deterministic RMSE may lead to suboptimal deployment decisions. In practice, grid operators must forecast load using imperfect weather inputs, and models that appear superior under perfect information may not deliver the most stable performance under realistic forecasting conditions.

5.3 Seasonal Patterns

Seasonal analysis reinforces these findings. During Winter, Linear and hybrid models perform similarly under observed temperatures, while XGBoost performs slightly better (10–20 MW lower RMSE), likely due to smoother, more predictable load patterns. In Summer, the hybrid model achieves the best baseline RMSE but degrades faster than Linear Regression at higher temperature uncertainty, mirroring top 10% load-hour behavior. XGBoost again underperforms in Summer, though the gap is smaller than for peak hours. This alignment occurs because most extreme load hours are concentrated in Summer, linking seasonal and peak-load performance patterns.

5.4 Hourly Performance

The heatmaps reveal a consistent intraday fragility pattern, with peak and daytime hours showing the largest RMSE increases as temperature uncertainty rises, while overnight periods remain comparatively stable across all model classes. Across models, baseline predictive performance is strongest during nighttime hours, when load variability is lower and less sensitive to temperature fluctuations.

In terms of model behavior, XGBoost slightly underperforms Linear Regression during overnight hours but performs competitively, and in some cases better, during mid-day periods when observed temperatures are used. As temperature uncertainty increases, the hybrid model exhibits pronounced degradation during high-variability mid-day and peak-load hours. XGBoost also degrades with increasing uncertainty, although its sensitivity is more spread across hours rather than concentrated only at peak periods.

Overall, rising temperature uncertainty weakens predictive performance most substantially during peak-demand hours, where load is strongly temperature-dependent. These results indicate that although more complex models can improve baseline accuracy, forecasting peak hours under uncertain inputs may require models that are explicitly designed for robustness rather than relying solely on increased flexibility.

6 Conclusion

This project developed a day-ahead load forecasting framework designed not only to maximize baseline accuracy but to evaluate model robustness under realistic temperature forecast uncertainty. By introducing Monte Carlo perturbations to temperature inputs, the analysis demonstrates that deterministic evaluation using observed temperatures can materially underestimate real-world forecasting error.

Under perfect temperature inputs, the hybrid Linear + XGBoost residual model achieved the strongest overall and peak-hour performance. However, as temperature uncertainty increased, model rankings shifted. The hybrid model degraded most rapidly, while Linear Regression exhibited greater stability at higher uncertainty levels. These results highlight a structural trade-off between predictive flexibility and robustness to exogenous input noise.

The findings emphasize that model selection for operational energy forecasting should incorporate robustness analysis in addition to baseline accuracy metrics. Forecast reliability during peak-demand periods, when system risk is highest, may require models that prioritize stability under uncertainty rather than purely minimizing deterministic RMSE.

Overall, this study illustrates how integrating uncertainty stress-testing into forecasting workflows can yield more deployment-relevant model evaluation and more informed model selection decisions.

7 Assumptions and Limitations

This analysis relies on several simplifying assumptions:

- The historical relationship between load and temperature remains stable over time, with no structural demand shifts.
- Temperature forecast errors are assumed to be unbiased and are simulated using a structured Monte Carlo scheme in which each synthetic forecast evolves from the previous forecast value, includes a partial correction toward the realized temperature, and incorporates an additive Gaussian shock with fixed standard deviation. This formulation preserves temporal correlation in forecast errors rather than assuming independent hour-to-hour perturbations.
- Forecast temperature bias is assumed constant across seasons and hours, with uncertainty represented solely through variance scaling.
- Only temperature uncertainty is modeled; other exogenous uncertainties (e.g., solar irradiance, behavioral demand shocks, outages) are excluded.
- Load is primarily driven by temperature and calendar structure, with no major policy or market changes during the evaluation period.
- Lagged load features (e.g., t24) are assumed to be known without error at prediction time, and no load forecast uncertainty is modeled.

While these assumptions allow clear isolation of temperature-driven robustness effects, real-world deployment would require ongoing retraining, regime-shift monitoring, and joint modeling of multiple uncertainty sources.